Solar Power Generation

Da Vinci Satellite

Issue:	1.0
Revision:	1.2
Revision Date:	28-04-2026
Author(s):	Nicolò Basso



VSV Leonardo Da Vinci

TU Delft

Solar Power Generation

Systems Team

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With the ESA Fly Your Satellite! 4 Programme

By

The Da Vinci Satellite Team

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Review and Approval

Title	Solar Power Generation Report		
Issue	1	**Revision Number**	1
Author(s)	Nicolò Basso	**Release Date**	26-04-2026
Review by	Stefano Speretta	**Date of Review**	27-04-2026
Approved by	TBD	**Date of approval**	TBD

Change Log

Reason for Change	**Author(s)**	**Issue**	**Revision**	**Date**
Add validation	Nicolò Basso	1	1	26-04-2026
Implement Professor Feedback	Nicolò Basso	1	2	28-04-2026

Change Record

Reason for Change	**Author(s)**	**Date**	**Section(s)**
Added validation of the code	Nicolò Basso	26-04-2026	3.2
Specified the changes in power due to degradation	Nicolò Basso	28-04-2026	2.2
Removed other averages for clarity	Nicolò Basso	28-04-2026	3.1
Added lower bounds of expected power for reference purposes	Nicolò Basso	28-04-2026	3.1

Distribution

This document is meant to be distributed within the DVS team and with the ESA FYS Team.

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Introduction 1

This report analyzes the power generation performance of the Da Vinci Satellite. While the spacecraft features an Attitude and Orbit Control Subsystem, the subsystem functionality is primarily dedicated to payload image acquisition and de-tumbling maneuvers within safe mode. Therefore, the satellite is modeled as being in an uncontrolled tumbling state throughout the majority of the mission.

The principal objective of this investigation is to establish a conservative baseline value for use in the Power Budget and Power Simulations Report. Prior assessments utilized an average power value derived from a random sampling of points on a sphere, which did not consider temperature fluctuations and end-of-life degradation. This updated report introduces a more comprehensive model that integrates thermal effects, lifetime performance loss, and a deterministic equal-area spherical sampling strategy.

Sun-incidence directions are sampled over the full unit sphere using a grid that is uniform in $\sin\theta$, ensuring equal solid-angle weighting without the statistical variability of random methods. For each sampled direction, the instantaneous electrical power output is evaluated accounting for cell temperature and radiation-induced degradation. The tumbling-average power is then computed as the arithmetic mean over all sampled directions.

1.1 Assumptions

The following assumptions have been adopted for this work:

- The rotation period is assumed to be significantly shorter than the duration of the sunlit orbital pass, a condition consistent with the expected tumbling rate of 1 to 10 degrees per second.
- The Earth albedo effect has been excluded from calculations, both for temperature effects and power generation. This leads to a conservative result.
- Internal heat dissipation from onboard components is neglected in the temperature estimation of the solar panels. The thermal balance of the panels is driven exclusively by incident solar radiation and infrared emissivity.
- A wiring efficiency of 98% has been considered.
- It has been considered that the solar cell will have a MPP tracker that allows the system to extract the maximum power available. The efficiency of the MPP derives from the ISIS Space Type B EPS, at around 87.5% at 5V.

This methodology ensures that the orbital power average provides a reliable lower bound for mission planning and system sizing.

Solar Cell Model

2

The DVS satellite is equipped with high-efficiency TJ 3G30-Advanced III–V triple junction Gallium Arsenide solar cells manufactured by AZUR SPACE.

To maintain accuracy throughout the mission, the model utilises linear interpolation between given data points to accounts for the degradation of maximum power point voltage and current as a function of electron fluence. As radiation exposure increases, the solar cell's internal structure degrades, leading to a drop in voltage and a slight increase in current. The following data was extracted from the datasheet. [1]

Table 2.1: Typical cell electrical performance.

Fluence (e/cm²)	BOL (0)	5E13	2.5E14	5E14	1E15	1E16
V_{mp} - [mV]	2411	2336	2298	2262	2230	2127
I_{mp} - [mA]	504	501	496	494	484	360

Table 2.2: Typical cell temperature coefficients.

Fluence (e/cm²)	BOL (0)	2.5E14	5E14	1E15
$\frac{dV_{mp}}{dT}$ - $\left[\frac{mV}{°C}\right]$	−6.7	−6.8	−7.1	−7.2
$\frac{dI_{mp}}{dT}$ - $\left[\frac{mA}{°C}\right]$	0.24	0.20	0.24	0.28

2.1 Configurations

Three main solar cells configurations have to be analysed in the simulations: a) all the solar panels working, b) −Z two solar panels removed, c) +Y two solar panels removed and d) two solar panels from +Y and −Z removed respectively. The reasons behind the configurations is explained in the Solar Panels Replacement and/or Removal Options Report [2]. This report, alongside the Power Simulations Report, has the scope to chose which of the four is the most adequate for the mission.

Table 2.3: a) All the panels working

side	+X	-X	+Y	-Y	+Z	-Z
# cells	4	4	4	4	2	2

Table 2.4: b) Two Z- panels lost

side	+X	-X	+Y	-Y	+Z	-Z
# cells	4	4	4	4	2	**0**

Table 2.5: c) Two Y+ panels lost

side	+X	-X	+Y	-Y	+Z	-Z
# cells	4	4	**2**	4	2	2

Table 2.6: d) Two Y+ and two −Z panels lost

side	+X	-X	+Y	-Y	+Z	-Z
# cells	4	4	**2**	4	2	**0**

2.2 Solar Cell Lifetime Degradation

A simulation on SPENVIS was executed for a orbit with $i = 97°$, $\Omega = 0$, $h = 600$ km and $e = 0$. The report can be accessed at [3]. The Cell settings are shown in Table 2.7.

Table 2.7: AZUR 3G30 key data.

Cell: Azur 3G30 (SR-NIEL 21eV)	
Electron/proton damage ratios:	
Pmax	1134.8
Voc	2864.6
Isc	613.4
Coverglass	
material	SiO2
density	2.32 (g/cm3)

With a mission lifetime of two years, the following Solar cell damage equivalent fluences were obtained (Table 2.8). The Total Fluence obtained through SPENNVIS is roughly 7% of the first provided data point in Table 2.1. Using a linear interpolation, the power decreases from 1.215 W to 1.211 W, a reduction of 0.33%.

Table 2.8: SPENVIS Fluences

Coverglass Thickness	Protons Fluence Φ_{max}	Electrons Fluence Φ_{max}	Total Fluence Φ_{max}
100 µm	3.484E+12	2.168E+11	3.706E+12

2.3 Power Model

In the calculations that follow, the temperature was kept into account using a linear model with the slopes taken directly from Table 2.2. The final value is a linear interpolation with the one at the BOL and the first data point. Note that the reference temperature is 28°C [1].

$$G_{\text{eff}} = G \cdot \cos\theta \tag{1.}$$

$$V_{\text{mp}_{\text{cell}}} = V_{\text{mp}_0} + \left.\frac{dV}{dT}\right|_{\Phi} (T_{\text{cell}} - T_{\text{ref}}) \tag{2.}$$

$$I_{\text{mp}_{\text{cell}}} = I_{\text{mp}_0} \frac{G_{\text{eff}}}{G_0} + \left.\frac{dI}{dT}\right|_{\Phi} (T_{\text{cell}} - T_{\text{ref}}) \tag{3.}$$

$$P_{\text{face}} = V_{\text{mp}_{\text{cell}}} \cdot I_{\text{mp}_{\text{cell}}} \cdot n_{\text{cells}} \cdot \eta_{\text{wiring}} \tag{4.}$$

Where a η_{wiring} of 0.98 was introduced to account for losses in wiring.

2.4 Temperature Effects

The temperature was obtained through a balance of heat flows for each face.

$$\dot{Q}_{\text{emitted}} = \dot{Q}_{\text{absorbed}} \tag{5.}$$

The emitted heat flow for each side is given by:

$$\dot{Q}_{\text{emitted}_i} = T^4 \cdot \sigma \cdot \varepsilon_{\text{coverglass}} \cdot A_i \tag{6.}$$

Where $\varepsilon_{\text{coverglass}} = 0.86$, as from [4] (CMG, AR coating).

The absorbed heat for face is given by:

$$\dot{Q}_{\text{absorbed}_i} = G \cdot A_i \cdot \alpha_{\text{solar-cell}} \cdot \cos\theta_i \tag{7.}$$

The albedo has been neglected, both for the temperature and for power generation. Therefore, when the face is not illuminated by the Sun, a temperature of 0 K has been given to the face as a placeholder, since in any case the solar cell is not generating power.

The final temperature for each cell is given by:

$$T = \sqrt[4]{\left(\frac{G \cdot \alpha_{\text{solar-cell}} \cdot \cos\theta_i}{\sigma \cdot \varepsilon_{\text{coverglass}}}\right)} \tag{8.}$$

Where an absorption coefficien of $\alpha_{\text{solar-cell}} = 0.91 \cdot 0.30 = 0.637$ has been used, which accounts for the $\sim 30\%$ electricity conversion. [1] Is it worth mentioning that the real temperature will be lower due to conduction with the aluminium structure, here neglected for simplicity purposes.

Following is the maximum temperature expected with the given model. Please acknowledge that, compared to data from satellite employing similar solar cells, this temperature is higher [5].

$$T_{\max} = \sqrt[4]{\left(\frac{1367\,\text{W m}^{-2} \cdot 0.637}{5.67 \cdot 10^{-8}\,\text{W m}^{-2}\text{K}^{-4} \cdot 0.86}\right)} = 365.7\,\text{K} = 92.7°\text{C} \tag{9.}$$

Figure 2.1 shows how the temperature of each face change with the direction of the Sun. As expected, the temperature gradient is the same for each face, since the temperature does not depend on the area, as can be seen in Eq. 8.

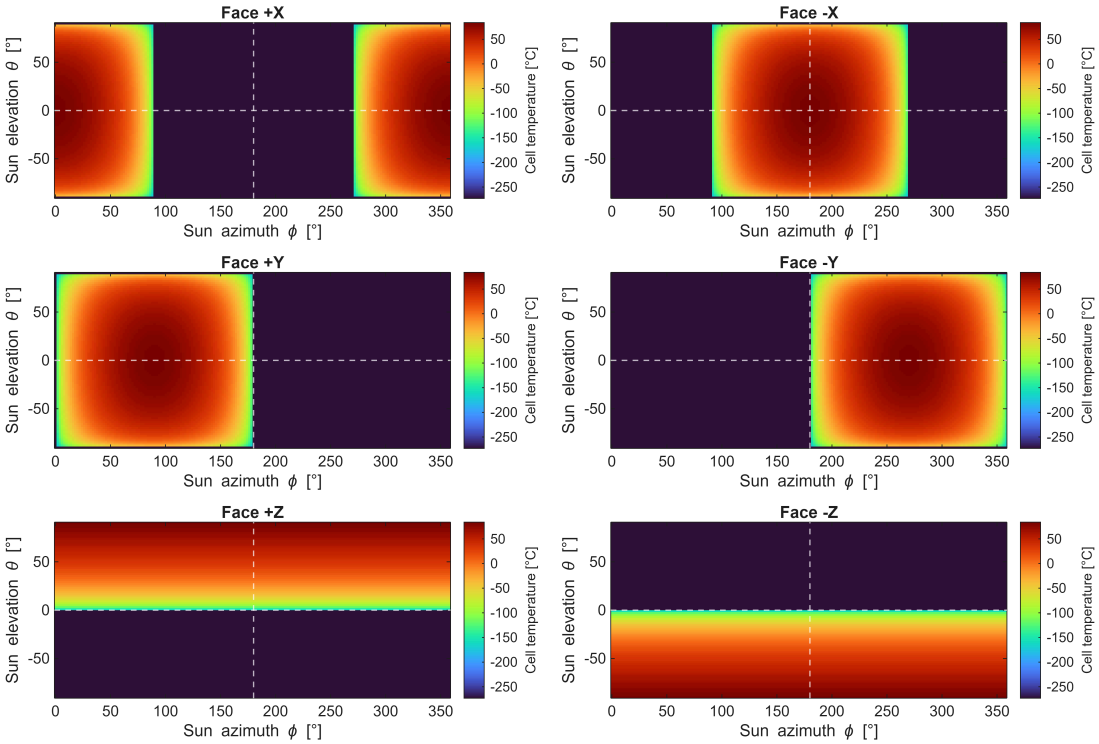


Figure 2.1: Temperature of each face with respect to the Sun direction.

2.5 Geometric Averaging Method

The average power generated during a tumbling motion is computed by uniformly sampling all possible sun-incidence directions over the unit sphere and averaging the resulting power output across all samples.

Sun directions are parameterised in spherical coordinates by elevation angle $\theta \in [-\pi/2, \pi/2]$ and azimuthal angle $\varphi \in [0, 2\pi)$. To ensure equal solid-angle weighting, the elevation grid is made uniform in $\sin\theta$ rather than in θ itself. The sample points are therefore defined as

$$\sin\theta_i = -1 + \frac{2(i-1)}{N_{\text{el}} - 1}, \quad i = 1, \ldots, N_{\text{el}} \tag{10.}$$

with N_{az} equally-spaced azimuth angles $\varphi_j = 2\pi j / N_{\text{az}}$, $j = 0, \ldots, N_{\text{az}} - 1$. Each pair (i, j) defines a unit sun vector in the spacecraft body frame:

$$\hat{s}_{ij} = \begin{pmatrix} \cos\theta_i \cos\varphi_j \\ \cos\theta_i \sin\varphi_j \\ \sin\theta_i \end{pmatrix} \tag{11.}$$

The instantaneous electrical power $P(\hat{s}_{ij})$ is evaluated for every sample point by calling `solar_power_output` with the corresponding sun vector. The tumbling-average power is then the arithmetic mean over all $N_{\text{el}} \times N_{\text{az}}$ directions:

$$\overline{P}_{\text{mean}} = \frac{1}{N_{\text{el}} N_{\text{az}}} \sum_{i=1}^{N_{\text{el}}} \sum_{j=1}^{N_{\text{az}}} P(\hat{s}_{ij}) \tag{12.}$$

Because the grid is uniform in solid angle, every sampled direction contributes equally and no additional weighting is required.

This approach differs fundamentally from an earlier method in which tumbling was simulated by composing successive quaternion rotations about a set of sampled spin axes. In that formulation, an initial attitude was rotated through a set of random angles drawn uniformly from $[0, 2\pi)$ for each of a distribution of rotation vectors generated via a Fibonacci sphere.

Results

3

The four configurations were evaluated in MATLAB using the geometric averaging method introduced above. Table 3.1 summarises the power output for the thermal, degradation-adjusted case together with the corresponding no-thermal BOL and EOL values, making it possible to isolate the effect of temperature and radiation losses across the different configurations.

As a reference, the no-thermal BOL case can be compared with the limit obtained from Cauchy's theorem. For any convex object tumbling randomly in 3D space, its average projected area is exactly 1/4 of its total surface area. Because solar power is directly proportional to the projected area facing the Sun, a tumbling satellite will therefore average about 1/4 of its total installed capacity in the idealised limit.

$$P_{\text{cell}_{\text{ideal}}} = V_{\text{mp0}} \cdot I_{\text{Imp0}} \approx 1.215 \text{ W} \tag{1.}$$

The theoretical ideal averages would be the total power divided by four. For the configuration A, which has a total of 20 solar cells:

$$P_{A_{\text{avg}}} = \frac{20 \cdot 1.215 \text{ W}}{4} = 6.075 \text{ W} \tag{2.}$$

Keeping into account the $\eta = 0.875 \cdot 0.98$, we obtain 5.21 W, which is close to the no-thermal BOL value of 5.191 W for configuration A that the code outputs. The remaining discrepancy is attributed to the discretisation of the surface grid.

3.1 Sunlit Tumbling Averages

The following set of results presented in Table 3.1 regard only the fully sunlit case, where the satellite is illuminated over the complete sphere of incidence directions. No eclipse has been kept into consideration: these values are the one meant to be fed into TUDat to obtain the average power generation value over an orbit.

Table 3.1: Sunlit tumbing average powers of different configurations.

Config	Min	Max	Avg
A	1.784 W	5.986 W	4.829 W
B	1.781 W	5.986 W	4.347 W
C	0.000 W	5.986 W	4.342 W
D	0.000 W	5.986 W	3.860 W

Figure 3.1 shows the distribution of power values and the mean for each configuration studied. Figure 3.2 presents the heatmap of the power generated of the different configuration with respect to the direction of the Sun vector. Configuration C and D show that there are some direction for which almost zero power is generated, due to the absence of panels on the -Z side.

For reference purpose only, using Eq. 3 at 180 km the maximum eclipse fraction is equal to 0.425 at the expected minima operational altitude. Therefore we can expect, as absolute lower bounds, the following values for each configuration: A) 2.776 W, B) 2.499 W, C) 2.496 W and D) 2.219 W.

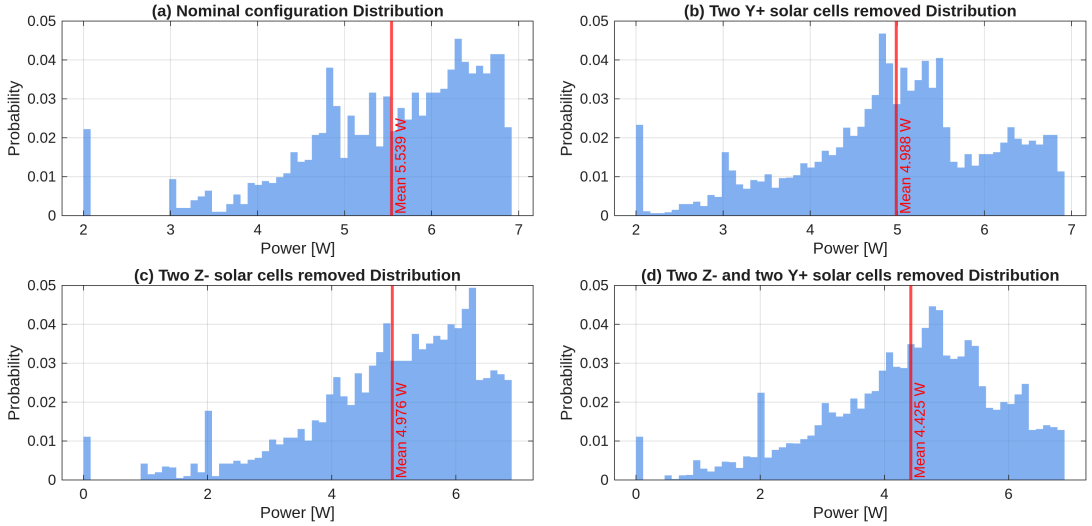


Figure 3.1: Distribution of sunlit power values across all sampled incidence directions.

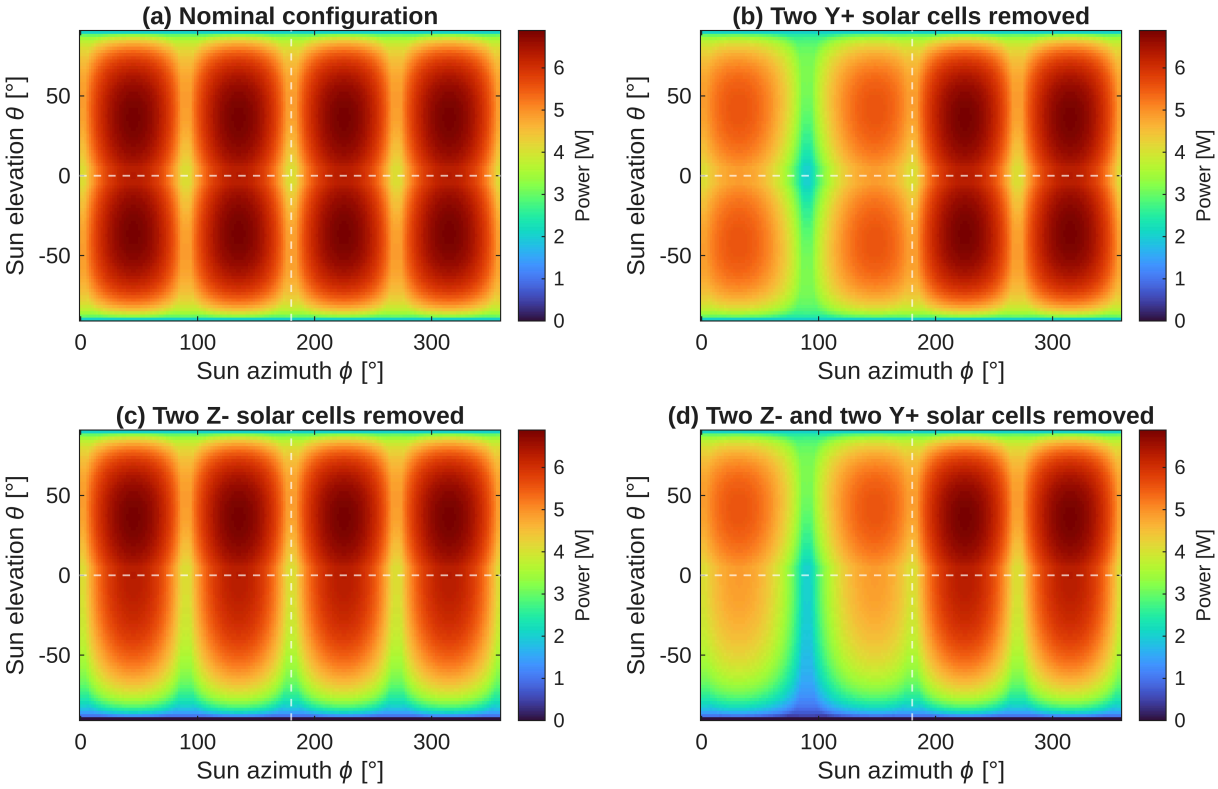


Figure 3.2: Heatmap of generated power over the Sun-direction sphere.

3.2 Validation

To validate the results obtained for the DVS configuration, a simulation was performed for the SPIRONE 2U Cubesats. This cubesat employs the same AZUR 3G30 solar cells, with a total of 19 cells (4 for +X, -X, +Y, -Y; 2 for +Z and 1 for -Z). The calculated generated power of the satellite was of 2.68 W. [6]. A configuration file was created on the software for this satellite. The main changes compared to DVS satellite are:

- The different number of cells;

- The electrical parameters of $I_{mp} = 502.9$ mA and $V_{mp} = 2409$ mV, provided in [6], differ slightly from the one in the datasheet.
- Degradation for the SPIRONE simulation was neglected.
- A lower efficiency of the MPPT of 85% was used, as given in [6].

Using these values, an average sunlit power of 4.5393 W was obtained. Utilising the following formula from literature to calculate the fraction of eclipse, with an orbit altitude of 600 km. [7]

$$f_{e_{max}} = \frac{1}{180°} \arccos\left(\frac{\sqrt{h^2 + 2hR}}{(R+h)}\right) = 0.367 \quad 3.$$

The final value power average obtained is of 2.816 W, 5% higher than the results presented in [6]. It is hence clear that, with a margin of 10%, a reasonable and conservative value for the DVS power budget can be obtained to cover this discrepancy.

References

[1] AZUR SPACE Solar Power GmbH, 3g30a Triple-Junction Solar Cell Assembly Datasheet (Tj3g30-Advanced 4x8), technical report, 2025.

[2] I. Gallo López, DVS Solar Panels Replacement and/or Removal Options, Internal Report No. DVS_SOLAR_PANEL_OPTIONS, 2026.

[3] Da Vinci Satellite Team, Solar Cell Damage Equivalent Fluences, Internal Report, 2025.

[4] Qioptiq, Space-Qualified Cover Glass Datasheet, (n.d.).

[5] A. Jara, P. Lepcha, S. Kim, H. Masui, T. Yamauchi, G. Maeda, and M. Cho, On-orbit electrical power system dataset of 1U CubeSat constellation, Data in Brief **45**, 108697 (2022).

[6] S. Gwon et al., *Design and Development of 2u SPIRONE Cubesat for Verification of Navigation Signal Generator in LEO Environment and Marine Plastic Observation*, https://doi.org/10.26077/WCGD-QJ38.

[7] H. Hakima and M. R. Emami, Deorbiter CubeSat System Engineering, The Journal of the Astronautical Sciences **67**, 1600 (2020).